

Nikos C. Baxevanis

Resume

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Summary. Pragmatic software engineer with two decades of experience writing and shipping software. Active in open source since 2011 and in security-focused engineering since 2017. Deep expertise in property-based testing, fuzzing, and program analysis; active in blockchain and smart contract security since 2017. Track record of building foundational test and analysis tools, driving major open-source projects, and finding critical vulnerabilities in production systems. Looking for roles where I can advance software assurance, mentor engineers, and push the boundaries of correctness tooling.

Key Accomplishments

- Designed and co-implemented the first fuzzing and model-based testing tools for the Stacks blockchain, including *rendezvous* and *madhouse-rs*, enabling consensus and smart contract validation in Rust.
- Helped shape and now maintain *Hedgehog*, a property-based testing framework for Haskell, F#, Scala, and R. In production use since 2016; originally powered *Echidna*, the first smart-contract fuzzer for Ethereum.
- Core maintainer and feature author for *AutoFixture*, a leading .NET testing library – contributed major sub-tools like *Fare*. Recognized by Mark Seemann.
- Held key engineering and technical leadership roles in two successful acquisitions—*KBMax* (2021) and *BetKerner* (2012)—with focus on correct, secure system development.

Professional Experience

Stacks Foundation. Remote. *2022—present. Resident.* Pioneered formal fuzzing and automated testing for blockchain smart contracts and consensus. Focus: critical-path security, differential testing, and building tools to help auditors and protocol developers. Joined via the Critical Bounties program.

Security Researcher & Consultant. Remote. *2017—present. Freelance.* Advise and audit DeFi projects on protocol correctness, testing, and secure smart contract engineering. Contributed input to Uniswap v3 liquidity strategy research led by Atis Elsts. Delivered support for fuzzing and adversarial testing.

KBMax. Remote. *2014—2021. Senior Software Engineer.* Principal developer for the first cloud-native CPQ with interactive 3D visualization on Azure. Led reliability and correctness work across full stack. Company acquired 2021.

BetKerner. Athens, Greece. *2009—2012. CTO.* Built and led a team for an automated betting analytics platform leveraging live data and expert signals. Contributed to product development through to acquisition in 2012.

Topologic. Athens, Greece. *2005—2009. Lead Software Engineer.* Shipped full-stack airport management and incident tracking software deployed at Athens Intl. Airport, using .NET & geospatial APIs.

Summary of Qualifications

- 20 years of software engineering; open source contributor since 2011.
- Expertise in property-based, model-based, fuzzing, and mutation testing.
- Deep experience in smart contract security and blockchain testing.
- Contributed to testing libraries and generators in Haskell, Rust, F#, and TypeScript.
- Advocates correctness-by-design, secure software practices, and mentoring.
- Known for uncovering subtle bugs in mission-critical systems.

Open Source & Writing. Curated technical posts, open-source work, and peer quotes available at nikosbaxevanis.com.

Education

Technological Educational Institute of Athens. 2002—2006. B.Sc. in Biomedical Engineering.

Logistics. Permanent resident in the Czech Republic. Greek citizen. Fluent in English and Greek.

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